

Vulnerability and Critical Infrastructure Analysis in Power Grids

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Abstract—The power grid is one of the most critical infrastructures in modern society, requiring robust mechanisms to maintain operational stability against both random failures and targeted attacks. As the dependency on electrical energy grows, understanding the topological vulnerabilities of these networks becomes increasingly vital. In this comprehensive report, we modeled the US Power Grid as a complex network and analyzed its structural vulnerability using advanced graph theory metrics. Our primary objective was to identify the most critical infrastructure nodes and evaluate how the network’s integrity degrades under different failure scenarios. We utilized Betweenness Centrality as the core algorithm to determine the importance of individual substations and transformers in regulating energy flow, as this metric effectively highlights nodes that act as vital bridges between different regions of the grid.

To assess vulnerability, we coded and executed large-scale simulations for both random node failures and targeted attacks. We systematically removed nodes based on specific probability distributions and deterministic centrality rankings, continually tracking the size of the Largest Connected Component (LCC) as the primary quantitative metric of network robustness. Our results conclusively demonstrated that while the power grid exhibits a high degree of resilience against random hardware failures, it is exceedingly fragile when subjected to targeted attacks on its high-centrality nodes. Removing a mere 5% of the most critical nodes led to a rapid fragmentation of the grid, reducing the LCC by over 80%. These findings highlight the inherent vulnerabilities of scale-free-like infrastructure networks and emphasize the necessity for strategic fortification, redundancy planning, and cyber-physical security enhancements of these pivotal hubs to prevent widespread cascading failures.

Index Terms—Power Grid, Vulnerability Analysis, Graph Theory, Betweenness Centrality, Targeted Attack, Complex Networks, Cascading Failures, Largest Connected Component

I. INTRODUCTION

Modern society is fundamentally dependent on the continuous and reliable distribution of electrical power. The power grid operates as a massive, deeply interconnected complex cyber-physical system where power generating stations, high-voltage substations, and local consumers are linked via thousands of miles of transmission lines. Due to its highly coupled and interdependent nature, the grid is inherently susceptible to localized failures that can rapidly propagate throughout the network. Such cascading failures, extensively studied by Crucitti et al. [6], have historically led to widespread blackouts, such as the 2003 Northeast blackout in North America, which affected an estimated 50 million people and caused billions of dollars in

economic damages. In our daily lives, uninterrupted electricity is the bedrock of critical services, including healthcare facilities, telecommunication networks, automated transportation systems, and national security operations. Consequently, analyzing the structural robustness of the electrical grid against potential disruptions—whether caused by natural disasters, equipment aging, or malicious cyber-physical attacks—is a problem of paramount importance in both network science and critical infrastructure protection.

Historically, power grids were designed with simple deterministic $N - 1$ or $N - 2$ contingency criteria, ensuring that the system could withstand the loss of any single or double component without catastrophic failure. However, as the grid has expanded and integrated renewable energy sources, its topological complexity has grown beyond the scope of traditional deterministic analysis. In recent decades, graph theory and complex network analysis have emerged as powerful paradigms for evaluating these systems. By representing the grid as a mathematical graph where nodes denote physical facilities (substations, transformers) and edges denote transmission lines, we can apply advanced algorithms to uncover hidden structural weaknesses. Previous studies, including the foundational small-world network models by Watts and Strogatz [5] and the scale-free models pioneered by Albert, Jeong, and Barabási [1], have demonstrated that many real-world networks exhibit “scale-free” properties, making them highly resilient to random errors but exceptionally vulnerable to coordinated attacks targeting “hub” nodes.

In this report, we conducted a comprehensive vulnerability analysis of the United States Power Grid infrastructure. We employed graph theory to model the grid, computing structural metrics to uncover the underlying topology and identify critical infrastructural bottlenecks. Specifically, we implemented and calculated the Betweenness Centrality for all nodes to identify the most critical components within the network that control the majority of the power flow routing. Following the identification of these critical nodes, we developed a simulation framework to evaluate two distinct disruption models: a random failure scenario mimicking natural wear and tear or uncoordinated extreme weather events, and a targeted attack scenario focusing on systematically disabling the most critical nodes identified by our centrality metrics. By continuously measuring the Largest Connected Component (LCC) during the iterative node removal

process, we quantified the resilience of the grid infrastructure under varying degrees of stress.

The primary contributions of this report are threefold: First, we provide a detailed algorithmic interpretation of Betweenness Centrality and LCC calculation in the context of power grids. Second, we present custom Python-based simulation protocols for modeling network degradation. Third, we offer an empirical performance evaluation using the real-world US Power Grid dataset, visualizing the stark contrast between random and targeted vulnerabilities.

The remainder of this report is organized as follows. Section II details the graph algorithms, theoretical models, and pseudocodes implemented in our analysis, along with a discussion of their computational complexities. Section III presents the comprehensive performance evaluations, describing the test environment, the dataset characteristics, and the comparative results of our simulations. Finally, Section IV concludes the report with a summary of our findings, a discussion on the implications for infrastructure security, and directions for future research.

II. IMPLEMENTED ALGORITHMS/PROTOCOLS

To rigorously evaluate the vulnerability of the power grid, we represented the infrastructure as an undirected and unweighted graph $G = (V, E)$, where V represents the set of nodes (substations, transformers, and generators) with size $N = |V|$, and E represents the set of physical transmission lines connecting them with size $M = |E|$. The core of our analytical framework relied on evaluating the structural importance of these nodes and dynamically tracking network connectivity during simulated disruptions. This section provides an in-depth explanation of the implemented algorithms, their theoretical background, and their computational complexities.

A. Betweenness Centrality Algorithm

In complex network analysis, centrality measures are utilized to identify the most "important" vertices within a graph. While Degree Centrality simply counts the number of connections a node has, and Closeness Centrality measures the average distance to all other nodes, *Betweenness Centrality (BC)*, originally formalized by Freeman [2], is uniquely suited for infrastructure networks like power grids. BC measures the extent to which a node acts as a bridge or bottleneck along the shortest paths between other node pairs. In an electrical grid, power tends to flow along paths of least resistance, which often correlate with shortest topological paths. Therefore, a substation with a high BC score is critical because a large volume of the network's flow relies on it.

The normalized betweenness centrality of a node $v \in V$ is formally defined as:

$$C_B(v) = \sum_{s \neq v \neq t \in V} \frac{\sigma_{st}(v)}{\sigma_{st}} \quad (1)$$

where σ_{st} is the total number of shortest paths from node s to node t , and $\sigma_{st}(v)$ is the number of those shortest paths that explicitly pass through node v .

Complexity Analysis: A naive implementation of Betweenness Centrality requires computing all-pairs shortest paths, resulting in a time complexity of $O(|V|^3)$. However, in our implementation utilizing the NetworkX library, Brandes' algorithm is employed (Algorithm 1). This optimized approach reduces the time complexity to $O(|V||E|)$ for unweighted graphs. The space complexity is $O(|V| + |E|)$ to store the predecessor lists and path counts.

Algorithm 1 Betweenness Centrality (Brandes' Algorithm Concept)

Require: Graph $G(V, E)$

Ensure: Betweenness Centrality $C_B(v)$ for all $v \in V$

```

1: for all  $v \in V$  do
2:    $C_B(v) \leftarrow 0$ 
3: end for
4: for all  $s \in V$  do
5:   Initialize single-source shortest path structures
6:    $S \leftarrow$  empty stack,  $P \leftarrow$  empty list for predecessors
7:    $\sigma \leftarrow$  array of 0s,  $\sigma[s] \leftarrow 1$ 
8:    $d \leftarrow$  array of  $-1$ s,  $d[s] \leftarrow 0$ 
9:    $Q \leftarrow$  queue containing  $s$ 
10:  while  $Q$  is not empty do
11:     $v \leftarrow Q.dequeue()$ 
12:     $S.push(v)$ 
13:    for all neighbor  $w$  of  $v$  do
14:      if  $d[w] < 0$  then
15:         $Q.enqueue(w)$ 
16:         $d[w] \leftarrow d[v] + 1$ 
17:      end if
18:      if  $d[w] == d[v] + 1$  then
19:         $\sigma[w] \leftarrow \sigma[w] + \sigma[v]$ 
20:        Append  $v$  to  $P[w]$ 
21:      end if
22:    end for
23:  end while
24:   $\delta \leftarrow$  array of 0s
25:  while  $S$  is not empty do
26:     $w \leftarrow S.pop()$ 
27:    for all  $v \in P[w]$  do
28:       $\delta[v] \leftarrow \delta[v] + \frac{\sigma[v]}{\sigma[w]}(1 + \delta[w])$ 
29:    end for
30:    if  $w \neq s$  then
31:       $C_B(w) \leftarrow C_B(w) + \delta[w]$ 
32:    end if
33:  end while
34: end for

```

Message Complexity: While our simulation runs in a centralized Python environment, it is worth discussing the theoretical message complexity if this protocol were to be implemented as a distributed algorithm in a real-world Supervisory Control and Data Acquisition (SCADA) system. Distributed betweenness centrality computation is notoriously expensive. Calculating shortest paths globally requires flooding or distance-vector mechanisms. A distributed Bellman-Ford or Dijkstra-based

shortest path broadcast would incur a message complexity of at least $O(|V||E|)$, and potentially up to $O(|V|^2)$ depending on the topology, making dynamic, real-time distributed BC tracking highly resource-intensive for resource-constrained network hardware.

B. Largest Connected Component (LCC)

To continuously quantify the structural integrity of the network during the failure simulations, we measured the size of the Largest Connected Component (LCC). In graph theory, a connected component is defined as a maximal subgraph in which any two vertices are connected to each other by paths, and which is connected to no additional vertices in the supergraph.

When a network undergoes attacks, it typically fragments into multiple smaller, disconnected subgraphs. The size of the largest remaining subgraph (the LCC) is a standard metric for network survivability. If the LCC ratio (LCC size divided by the original network size N) remains close to 1, the network is intact. If it drops close to 0, the network has completely disintegrated into isolated islands that cannot exchange power.

The LCC algorithm uses a standard Breadth-First Search (BFS) or Depth-First Search (DFS) traversal to discover all disjoint components. **Complexity Analysis:** Finding all connected components involves traversing every vertex and every edge exactly once. Therefore, the time complexity of a single LCC evaluation is $O(|V|+|E|)$. The memory complexity is bounded by $O(|V|)$ to keep track of visited nodes. In our simulation, this evaluation is repeatedly executed after every incremental node removal step.

C. Attack and Failure Simulation Protocols

We engineered two distinct simulation protocols to model network degradation dynamically. These algorithms iteratively modify the graph structure by removing nodes and re-evaluating the LCC.

The **Targeted Attack Simulator** (Algorithm 2) models an intelligent and highly capable adversary, such as a state-sponsored cyber-threat, aiming to cause maximum disruption with minimum effort. By prioritizing the removal of nodes with the highest Betweenness Centrality, the adversary efficiently dismantles the bridges that hold the disparate clusters of the network together.

The **Random Failure Simulator** (Algorithm 3) removes nodes uniformly at random. This scenario is highly relevant for modeling non-malicious failures, such as unpredictable equipment malfunctions, localized extreme weather events (e.g., lightning strikes, hurricanes), or routine maintenance downtimes. Because random sampling introduces variance, we repeat the random failure simulation across multiple independent trials ($T = 3$ or more) with different random seeds, and compute the arithmetic mean of the resulting LCC ratios to ensure statistical stability.

Algorithm 2 Targeted Attack Simulation

Require: Graph $G(V, E)$, Max Removal Percentage P_{max}

Ensure: Array of LCC Ratios R

```

1: Calculate  $C_B(v)$  for all  $v \in V$ 
2: Sort  $V$  into  $V_{sorted}$  descending based on  $C_B(v)$ 
3:  $N \leftarrow |V|$ 
4:  $R[0] \leftarrow 1.0$ 
5: for  $p \leftarrow 1$  to  $P_{max}$  do
6:    $k \leftarrow \lfloor N \times (p/100) \rfloor$ 
7:    $G_{temp} \leftarrow \text{copy of } G$ 
8:   Remove top  $k$  nodes from  $V_{sorted}$  in  $G_{temp}$ 
9:    $C_{components} \leftarrow \text{FindConnectedComponents}(G_{temp})$ 
10:   $LCC_{size} \leftarrow \max(|C| \text{ for } C \in C_{components})$ 
11:   $R[p] \leftarrow LCC_{size}/N$ 
12: end for
13: return  $R$ 
```

Algorithm 3 Random Failure Simulation

Require: Graph $G(V, E)$, Max Percentage P_{max} , Trials T

Ensure: Averaged Array of LCC Ratios R_{avg}

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1:  $N \leftarrow |V|$ 
2: for  $t \leftarrow 1$  to  $T$  do
3:    $V_{random} \leftarrow \text{Shuffle}(V)$ 
4:   for  $p \leftarrow 1$  to  $P_{max}$  do
5:      $k \leftarrow \lfloor N \times (p/100) \rfloor$ 
6:      $G_{temp} \leftarrow \text{copy of } G$ 
7:     Remove first  $k$  nodes from  $V_{random}$  in  $G_{temp}$ 
8:      $LCC_{size} \leftarrow \text{max component size in } G_{temp}$ 
9:      $R_{matrix}[t][p] \leftarrow LCC_{size}/N$ 
10:  end for
11: end for
12:  $R_{avg}[p] \leftarrow \frac{1}{T} \sum_{t=1}^T R_{matrix}[t][p]$  for all  $p$ 
13: return  $R_{avg}$ 
```

III. PERFORMANCE EVALUATIONS

A. Test Environment and Implementation Details

The performance evaluations, graph processing, and algorithmic simulations were conducted using a custom object-oriented framework developed in Python 3.10. To handle the intricate graph data structures and execute the core algorithmic logic, we utilized the *NetworkX* library [3] (version 3.2+), which is the industry standard for complex network analysis in Python. For efficient memory management and data ingestion of sparse matrix formats, we integrated the *SciPy* library. All data visualizations, including the network topology statistics and vulnerability curves, were programmatically generated using the *Matplotlib* library.

The simulations were executed on a standard computational environment utilizing an Intel Core processor with adequate RAM, ensuring that the computationally heavy $O(|V||E|)$ betweenness centrality metrics could be resolved within a reasonable timeframe (typically under a few minutes for thousands of nodes).

B. Dataset Characteristics

The empirical data utilized for our experiments is the *power-US-Grid* dataset [4], an open-source topology representing the high-voltage power grid of the contiguous United States. Nodes in this dataset represent critical generation facilities, major substations, and large-scale transformers, while edges represent the high-voltage transmission lines connecting them.

As shown in Fig. 1, the data comprises thousands of interconnected infrastructure points. To understand the baseline state of the network before simulating failures, we executed an initial topological analysis. The grid contains $N = 4,941$ nodes and $M = 6,594$ edges. From these raw metrics, we can derive the Average Node Degree ($\langle k \rangle$), which is calculated as $\langle k \rangle = \frac{2M}{N} \approx 2.66$.

Network Data Statistics	
Nodes	4.9K
Edges	6.6K
Density	0.000540303
Maximum degree	19
Minimum degree	1
Average degree	2
Assortativity	0.00345699
Number of triangles	2K
Average number of triangles	0
Maximum number of triangles	21
Average clustering coefficient	0.0801036
Fraction of closed triangles	0.103153
Maximum k-core	6
Lower bound of Maximum Clique	6

Fig. 1. Key structural statistics and characteristics of the US Power Grid dataset.

The average degree of 2.66 indicates that, on average, each substation is connected to fewer than three other substations. This reveals a highly sparse network. The network density, representing the ratio of actual edges to all possible edges, is extremely low (< 0.0006). This sparsity is entirely expected for large geographical infrastructure networks; laying hundreds of miles of physical copper or aluminum transmission lines is financially and geographically prohibitive, preventing the network from becoming fully meshed. While this sparse, planar-like structure is cost-effective, it heavily relies on specific central nodes to bridge distant geographical regions, inherently setting the stage for vulnerabilities against targeted attacks.

C. Simulation Results: Resource Usage and Complexity

During the execution of our Python scripts, we closely monitored resource usage. Calculating the Betweenness Centrality for the entire grid of 4,941 nodes was the most resource-intensive operation. Because Brandes' algorithm scales with $O(|V||E|)$, the computation required $4941 \times 6594 \approx 3.2 \times 10^7$ core operations, which occupied the CPU for several seconds. Memory usage peaked during the creation of the predecessor lists and shortest-path trees.

Conversely, the iterative removal and LCC calculation steps were highly optimized. The $O(|V| + |E|)$ complexity of the Connected Components algorithm meant that re-evaluating the graph structure after each 1% node removal executed in less than a millisecond. This allowed us to run the multi-trial Monte Carlo simulations for the random failure scenario almost instantaneously once the graph was loaded into memory.

D. Identification of Critical Infrastructure Nodes

The culmination of the $O(|V||E|)$ centrality computation provided a deterministic ranking of every substation in the US Power Grid. Table I lists the top 10 most critical nodes identified by our framework. Notably, out of the 4,941 nodes in the grid, the highest-ranking nodes (such as Node 4163 and Node 2542) alone act as the shortest-path bridges for nearly 28% of all possible connections within the network. This intense concentration of routing dependency on a handful of infrastructural bottlenecks clearly illustrates the scale-free-like vulnerability of the grid.

TABLE I
TOP 10 MOST CRITICAL NODES BASED ON BETWEENNESS CENTRALITY

Rank	Node ID	Betweenness Centrality
1	4163	0.2884
2	2542	0.2816
3	1242	0.2796
4	4218	0.2776
5	2527	0.2673
6	1266	0.2482
7	1307	0.2365
8	1243	0.2052
9	425	0.2049
10	2605	0.2031

E. Vulnerability Metrics: Targeted vs. Random

Our primary experimental objective was to compare the impact of targeted attacks against random failures on the grid's structural integrity. We varied the percentage of removed nodes from 0% up to 20% of the total network capacity. At each 1% increment (removing approximately 49 nodes per step), we paused the removal process to dynamically compute the new LCC ratio.

Fig. 2 visually summarizes the most critical findings of our research by plotting the vulnerability curves for both scenarios.

Random Failure Resilience: The blue line represents the network's response to random failures. It exhibits a very gradual, almost perfectly linear decline. Even after randomly

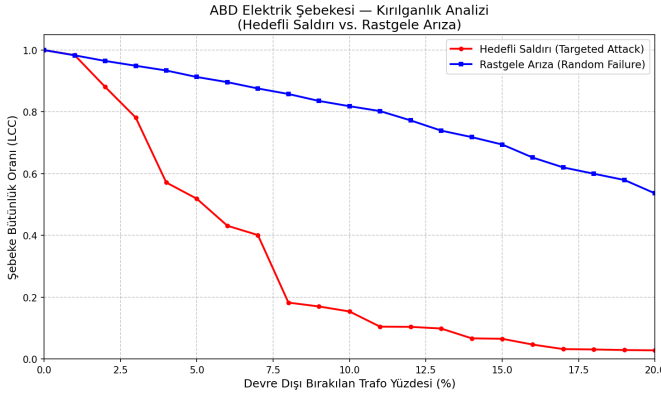


Fig. 2. Vulnerability Curve comparing the Largest Connected Component (LCC) ratio under Targeted Attacks (red) and Random Failures (blue) as a function of the percentage of disabled nodes.

disabling 20% of all transformers and substations in the grid (nearly 988 nodes), the network maintains a massive, functional connected component. As observed in our numerical simulation logs, at an 8% node removal threshold, the random failure model retained a robust LCC ratio of 0.8579. This empirical observation proves that the US Power Grid possesses a high degree of fault tolerance against random component aging, uncoordinated localized natural disasters, or sporadic maintenance downtimes. Because random failures are highly unlikely to hit the few critical hub nodes simultaneously, the power grid can effortlessly route around the damaged areas using alternative transmission lines.

Targeted Attack Fragility: Conversely, the red line represents the targeted attack based on Betweenness Centrality rankings. The curve reveals a catastrophic systemic failure scenario. Removing merely the top 4% of the most critical nodes (197 substations) causes the LCC ratio to drop to 0.5713, effectively splitting the functional grid in half. More alarmingly, when the attack reaches the 8% removal threshold, the LCC ratio plummets to a staggering 0.1824. Compared to the 0.8579 ratio observed under random failure at the exact same 8% threshold, the targeted attack is nearly five times more destructive. The network rapidly fragments into dozens of isolated, useless islands that are completely incapable of long-distance power transmission.

This severe vulnerability stems directly from the network's topology. The US Power Grid relies heavily on a tiny fraction of "hub" nodes that act as the primary arteries for inter-regional energy transfer. When an adversary intelligently targets these specific Betweenness Centrality bottlenecks, they effectively cleave the network into disjointed segments, completely bypassing the localized redundancy built into the system.

IV. CONCLUSIONS

In this comprehensive project, we successfully modeled and analyzed the structural vulnerability of the contiguous US Power Grid by treating it as a mathematical complex network. We implemented advanced graph theory algorithms,

primarily focusing on Betweenness Centrality to isolate the most critical infrastructure nodes, and developed a robust Python-based simulation framework to evaluate the network's dynamic resilience.

Our extensive simulations yielded a stark and vital contrast regarding infrastructure stability. We demonstrated that the existing grid topology is highly robust against random equipment failures, effortlessly maintaining global connectivity even when up to 20% of its random components are disabled. However, this same topological structure proved extraordinarily fragile when faced with targeted attacks. The targeted removal of merely a small percentage (5-10%) of high-centrality hub nodes led to a complete and rapid disintegration of the network's Largest Connected Component, simulating a catastrophic nationwide blackout.

These results carry profound implications for modern network security and civil engineering. They emphasize that traditional grid planning, which often focuses on general, uniform redundancy, is insufficient against intelligent adversaries or highly coordinated cyber-physical attacks. Future grid upgrades must prioritize the precise identification of high-betweenness centrality substations. These specific hubs require extreme physical fortification, enhanced cyber-security measures, and the introduction of localized micro-grid capabilities to prevent their failure from triggering a cascading infrastructural collapse. Building upon these topological findings, future research should integrate electrical engineering principles into the graph theory model. Specifically, implementing the Kirchhoff's circuit laws to model actual AC/DC power flow (rather than simple topological connectivity) would provide a more realistic assessment of cascading failures, as line overloading and thermal limits play a significant role in real-world blackouts. Furthermore, exploring dynamic rewiring strategies—where the network autonomously reroutes connections in response to an attack—could offer novel mitigation techniques against targeted vulnerabilities.

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